

Ramcharan Rajpurohit

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GitHub | LinkedIn | Website | Codolio

EDUCATION

B.Tech, IIT Jodhpur	CGPA: 7.19/10	Nov 2023 – May 2027
Senior Secondary, RBSE Board	Grade: 98.20%	Apr 2020 – May 2022

WORK EXPERIENCE

- **AI Engineering Intern at PowersMy.biz** [\[Live Platform\]](#) | Oct 2025 – Present
 - Built a production AI assessment platform for JEE and NEET coaching institutes using FastAPI, LangGraph, React, Flutter, MongoDB, Redis, and GCP, supporting **1,000+ concurrent exam-takers**.
 - Optimized high-concurrency exam APIs with **Redis caching**, atomic MongoDB operations, and off-thread bcrypt processing to sustain peak load.
 - Engineered a multimodal **AI evaluation pipeline** using vision models to grade handwritten subjective answers across 8 question formats and identify learning gaps.
 - Designed an **asynchronous grading queue** across 2 services that returns objective scores instantly while the pipeline evaluates written submissions in the background.
 - Collaborated across **5 production repositories** (backend, web, mobile, and AI-agent) via Git and GitHub, raising pull requests, reviewing code, resolving merge conflicts, and writing test cases to ship features safely.
 - Led deployment of the FastAPI backend on **Google Cloud Platform** with Cloud Run, VPC connector, auto-scaling, and concurrency tuning to handle exam-day traffic spikes.

PROJECTS

- **DSA Visualizer** – Interactive Algorithm Learning Platform [\[Live Site\]](#) | [\[GitHub\]](#)
 - Developed a framework-free algorithm visualizer in raw ES6 and HTML5 Canvas, animating 15+ sorting, graph, and tree algorithms step by step.
 - Architected a class-based OOP rendering engine where each algorithm runs as a self-contained plug-in subclass.
 - Implemented frame-by-frame execution controls, including recursion tracing and backward stepping through each visualized algorithm.
- **DevPlaza** – Competitive Programming Profile Aggregator [\[Live Site\]](#) | [\[GitHub\]](#)
 - Consolidated programming stats from 5 coding platforms (Codeforces, LeetCode, GeeksforGeeks, CodeChef, AtCoder) into one MERN dashboard.
 - Integrated private GraphQL and REST endpoints using rotating user agents and custom headers to pull undocumented stats.
 - Hardened every platform feed with retry logic and fallback parsers to handle upstream schema changes and outages.
 - Secured the app with a unified authentication layer of 3 sign-in methods: JWT sessions, Google OAuth, and email OTP.

TECHNICAL SKILLS

- **AI/ML & GenAI:** LangGraph, LangChain, Gemini API, OpenAI API, Vision Models, Prompt Engineering
- **Programming Languages:** C++, C, Python, JavaScript, TypeScript
- **Frontend Development:** React.js, HTML5, CSS3, TailwindCSS, Redux, Flutter
- **Backend Development:** Node.js, Express.js, FastAPI, REST APIs, GraphQL, JWT, OAuth 2.0, MongoDB, Redis
- **Cloud & Tools:** GCP (Cloud Run, VPC), Docker, GitHub Actions, Git, Postman, Linux, Vercel, CI/CD

ACHIEVEMENTS

- Ranked **AIR 12620** in **JEE Advanced 2023** among India's top engineering aspirants.
- Selected for **Amazon Summer School 2025**.
- Achieved **2nd District Rank** in Class 12 RBSE Board Exams with a 98.20% aggregate.
- Solved **350+ algorithmic problems** across LeetCode, CodeChef, and Codeforces.

LEADERSHIP

- Coordinated events and creative direction as Core Member of Dramatics Society (**Dramebazz**), IIT Jodhpur.
- Represented IIT Jodhpur at **Inter IIT Cultural Meet 7.0**, directing stage productions and mime performances.